

TMR 4190 FINITE ELEMENT METHODS IN STRUCTURAL ANALYSIS

Autumn 2026

EXERCISE 2

Problem 1

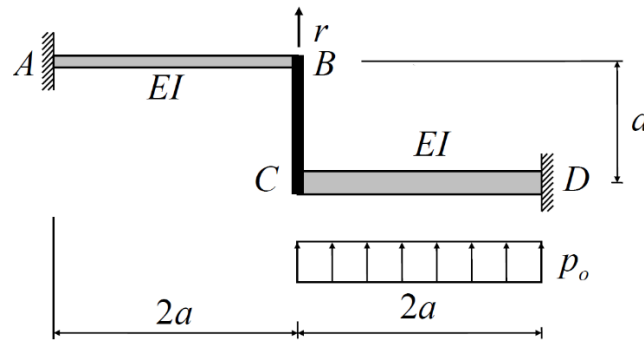


Figure 1

Figure 1 shows geometry, stiffness, loading and boundary conditions for a plane beam structure consisting of two horizontal beams AB and CD connected to each other through a rigid link BC . Only bending stiffness should be taken into account, i.e. axial and shear deformations should be neglected.

- Establish the stiffness relation $\mathbf{K}\mathbf{r} = \mathbf{R}$ for a minimum number of degrees of freedom, and determine $\mathbf{r}$.
- Calculate and sketch the moment and shear force distribution for the two beams AB and CD .

Problem 2

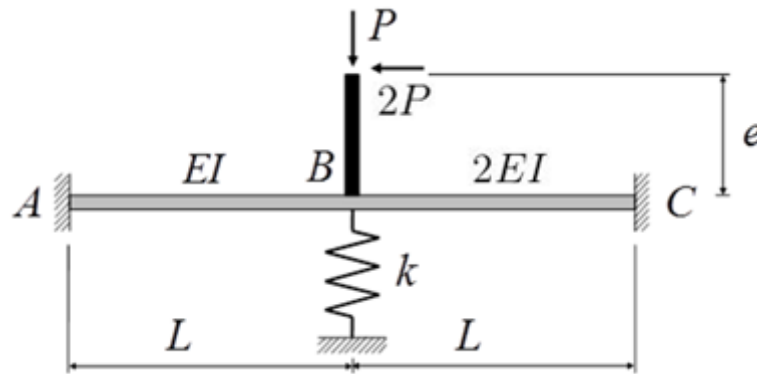


Figure 2

Figure 2 shows a plane beam, fixed at A and C , with length $2L$ and bending stiffness EI and $2EI$. The beam is supported by a vertical elastic spring with stiffness k at point B , and the loading consisting of a horizontal and vertical force (respectively $2P$ and P), is transferred to the beam through a vertical rigid link with length $e = L/12$, that is rigidly connected to the beam at point B . Only bending stiffness shall be taken into account, i.e. axial and shear deformations should be neglected.

- Establish the stiffness relation $\mathbf{K}\mathbf{r} = \mathbf{R}$ for a minimum number of degrees of freedom
- Determine $\mathbf{r}$ when the spring stiffness is $k = 0$.

Problem 3

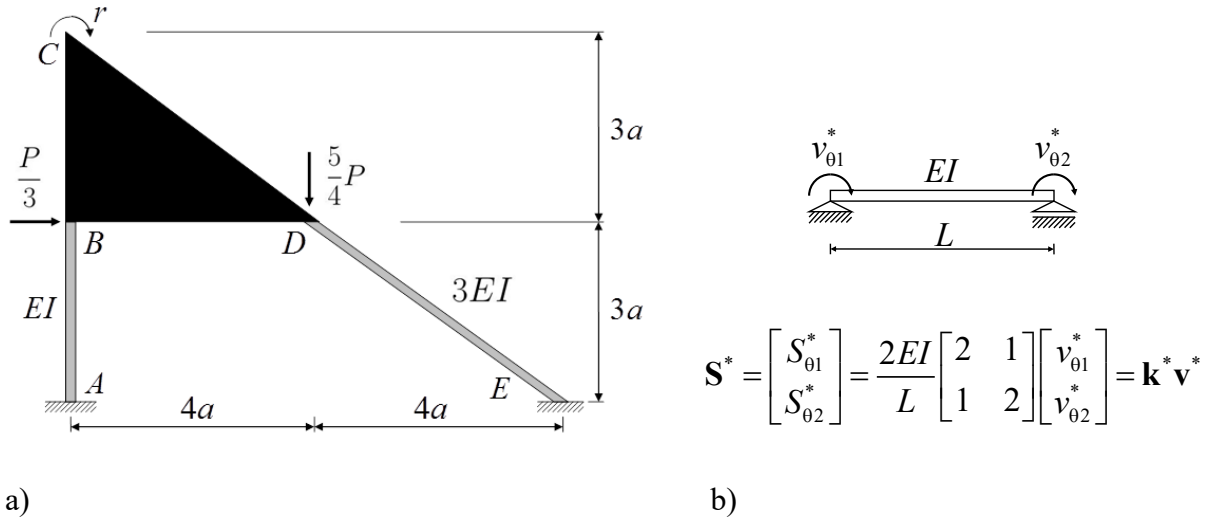


Figure 3

Figure 3a shows a plane beam structure consisting of two beam elements AB and DE rigidly connected to a rigid panel BCD at B and D , and fixed at A and E , respectively. Beam AB have length $3a$ and bending stiffness EI , and beam DE have length $5a$ and bending stiffness $3EI$, respectively. A horizontal concentrated load $P/3$ acts at point B and a downward concentrated load $5P/4$ acts at point D , as shown in Figure 2a. Only bending stiffness shall be taken into account, i.e. axial and shear deformations should be neglected.

Since the axial deformations are neglected, the deformation of the beam structure will be completely defined by one single degree of freedom (dof), e.g. the rotation about point C , denoted r .

- Based on the beam element in Figure 3b establish the topology matrix, $\mathbf{a}^*$, relating the element dofs, $\mathbf{v}^* = [v_{\theta 1}^* \ v_{\theta 2}^*]^T$, for beam AB and DE , to $\mathbf{r} = [r]$, respectively.
- Determine the stiffness matrix $\mathbf{K}$ corresponding to $\mathbf{r}$.
- Establish the load vector $\mathbf{R}$ due to the horizontal load $P/3$ acting at point B and the downward load $5P/4$ acting at point D , and solve for the unknown nodal dof $\mathbf{r}$.
- Calculate and sketch the moment distribution for beam AB and DE , respectively.

Appendix: Stiffness matrices and consistent load vectors for 3 and 4 d.o.f. beam elements

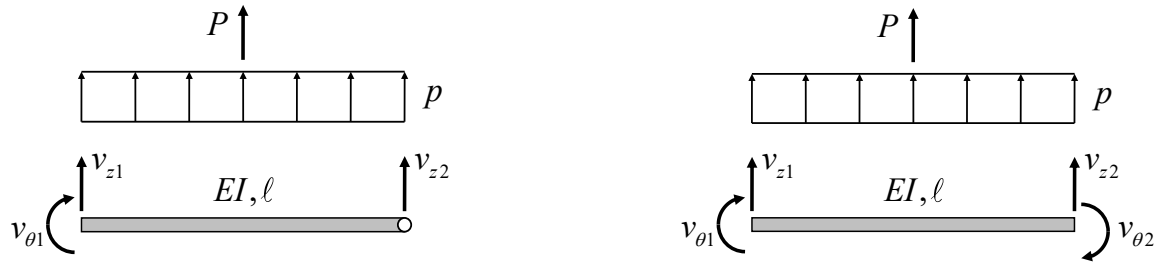


Figure 3: a) 3 d.o.f. beam element, and b) 4 d.o.f. beam element

Element stiffness matrices and consistent load vectors for the two beam elements shown in Figure 3. The element stiffness matrices account for bending deformations, only.

3 d.o.f. beam element:

$$\mathbf{S} = \begin{bmatrix} S_{z1} \\ S_{\theta1} \\ S_{z2} \end{bmatrix} = \frac{3EI}{\ell^3} \begin{bmatrix} 1 & -\ell & -1 \\ -\ell & \ell^2 & \ell \\ -1 & \ell & 1 \end{bmatrix} \begin{bmatrix} v_{z1} \\ v_{\theta1} \\ v_{z2} \end{bmatrix} + \frac{p\ell}{8} \begin{bmatrix} -5 \\ \ell \\ -3 \end{bmatrix} + \frac{P}{16} \begin{bmatrix} -11 \\ 3\ell \\ -5 \end{bmatrix} = \mathbf{kv} + \mathbf{S}^o$$

4 d.o.f. beam element:

$$\mathbf{S} = \begin{bmatrix} S_{z1} \\ S_{\theta1} \\ S_{z2} \\ S_{\theta2} \end{bmatrix} = \frac{2EI}{\ell^3} \begin{bmatrix} 6 & -3\ell & -6 & -3\ell \\ -3\ell & 2\ell^2 & 3\ell & \ell^2 \\ -6 & 3\ell & 6 & 3\ell \\ -3\ell & \ell^2 & 3\ell & 2\ell^2 \end{bmatrix} \begin{bmatrix} v_{z1} \\ v_{\theta1} \\ v_{z2} \\ v_{\theta2} \end{bmatrix} + \frac{p\ell}{12} \begin{bmatrix} -6 \\ \ell \\ -6 \\ -\ell \end{bmatrix} + \frac{P}{8} \begin{bmatrix} -4 \\ \ell \\ -4 \\ -\ell \end{bmatrix} = \mathbf{kv} + \mathbf{S}^o$$